# IGCSE 0607 Practice Questions — E1.16 — Combined Questions & Markscheme

## Exponential growth and decay

**Total Questions:** 12 — **Total Marks:** 63 Marks

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**1** A population of bacteria increases exponentially at a constant rate of $p\%$ per hour.
At 09 : 00, the population is 8 000.
At 12 : 00 on the same day, the population is 27 000.

**(a)** **(a)** Work out the value of $p$.

$p =$ ..... **[3]**

**(b)** Work out the population of the bacteria at 14 : 00 on the same day.

..... **[2]**

**2** A rare stamp is bought for \$450. Its value increases exponentially at a rate of 6.5% per year. Calculate the value of the stamp after 12 years.

\$ ..... **[3]**

**3** A machine is bought for \$32 000. It depreciates exponentially at a rate of 12% each year.

Calculate the value of the machine after 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

**4** The population of a town is 65000.
The population increases exponentially at a rate of 2.8% per year.

**(a)** Calculate the expected population of the town after 5 years.
Give your answer correct to the nearest integer.

..... **[2]**

**(b)** Calculate the overall percentage increase in the population over the 5 years.
Give your answer correct to 1 decimal place.

..... % **[2]**

**5** The population of a town was 85 000 on 1 January 2010.
The population increases exponentially at a constant rate of $r\%$ per year.
On 1 January 2022, the population was 112 000.

**(a)** **(a)** Calculate the value of $r$, correct to 3 significant figures.

$r =$ ..... **[3]**

**(b)** Assuming the rate of increase remains the same, calculate the expected population on 1 January 2030.
Give your answer correct to the nearest thousand.

..... **[2]**

**6** A machine costs \$45000. Its value depreciates exponentially at a rate of $x\%$ per year. After 4 years, the value of the machine is \$26000.

**(a)** Calculate the value of $x$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

**(b)** The machine will be replaced when its value first falls below \$10000.
Find the number of complete years before the machine is replaced.

..... years **[3]**

**7** Consider the depreciation of two different vehicles over time.

**(a)** A car is purchased for \$28 500. Its value depreciates exponentially at a rate of 12% per year.
Calculate the value of the car after 5 years, giving your answer correct to the nearest dollar.

\$ ..... **[3]**

**(b)** A van, initially purchased for \$35 000, depreciates exponentially to a value of \$15 200 after 4 years.
Calculate the annual percentage rate of depreciation for this van.

..... % **[4]**

**8** The mass, $M$ grams, of a radioactive substance decays exponentially at a rate of $r\%$ per day.
After 2 days, the mass is 80 grams.
After 5 days, the mass is 40.96 grams.

**(a)** Calculate the value of $r$.

$r =$ ..... **[3]**

**(b)** Find the initial mass of the substance.

..... grams **[2]**

**(c)** Calculate the number of complete days from the start when the mass is first less than 10 grams.

..... days **[3]**

**9** The population of a species of insect increases exponentially.
Initially, there are 2500 insects.
After 6 weeks, there are 9100 insects.

**(a)** Calculate the percentage increase in the population per week.

..... % **[4]**

**(b)** Calculate the insect population after 10 weeks, giving your answer correct to the nearest 100.

..... **[2]**

**(c)** Find the number of complete weeks it takes for the initial population to increase to over 50 000.

..... weeks **[3]**

**10** A new car is bought for \$26000. The value of the car depreciates exponentially at a rate of 15% each year.

Work out the value of the car after 4 years.
Give your answer correct to the nearest dollar.

% ..... **[3]**

**11** A population of bacteria in a dish grows exponentially.

**(a)** Initially, there are 1500 bacteria in the dish.
The population increases by 8% every hour.
Work out the population of bacteria after 12 hours.
Give your answer correct to the nearest whole number.

..... **[2]**

**(b)** Work out the least number of complete hours it will take for the initial population of 1500 bacteria to be greater than 3000.

..... hours **[3]**

**12** **(a)** The value of a painting increases exponentially by $r\%$ each year.
After 2 years, the total percentage increase in the value of the painting is 44%.
Work out the value of $r$.

$r =$ ..... **[3]**

**(b)** The value of a rare stamp increases exponentially.
Its value doubles every 15 years.
The current value of the stamp is \$80.

Work out the value of the stamp after 45 years.

% ..... **[2]**

**1** **(a)** **(a)** $8000(1+\frac{p}{100})^3 = 27000$ **[M1]**

$1+\frac{p}{100} = \sqrt[3]{\frac{27000}{8000}} = \frac{3}{2}$ **[M1]**

50 **[A1]**

**(b)** $27000 \times (1.5)^2$ or $8000 \times (1.5)^5$ **[M1]**

60 750 **[A1]**

[Back to Question] *[Total Marks: 5]*

**2** $450 \times \left(1+\frac{6.5}{100}\right)^{12}$ **[M1]**

$450 \times 1.065^{12}$ **[M1]**

958.01 (accept 958) **[A1]**

[Back to Question] *[Total Marks: 3]*

**3** $32000 \times (1-0.12)^5$ oe **[M2]**

For $32000 \times 0.88^n$ where $n > 1$, award **[M1]**

16887 **[A1]**

[Back to Question] *[Total Marks: 3]*

**4** **(a)** $65000 \times \left(1+\frac{2.8}{100}\right)^5$ or $65000 \times 1.028^5$ **[M1]**

74624 **[A1]**

**(b)** $\frac{74624-65000}{65000} \times 100$ or $(1.028^5 - 1) \times 100$ **[M1]**

14.8 **[A1]**

[Back to Question] *[Total Marks: 4]*

**5** **(a)** **(a)** $85000(1+\frac{r}{100})^{12} = 112000$ **[M1]**

$1+\frac{r}{100} = \left(\frac{112}{85}\right)^{\frac{1}{12}}$ **[M1]**

2.33 (from 2.327...) **[A1]**

**(b)** $85000 \times \left(1+\frac{2.327...}{100}\right)^{20}$ or $112000 \times \left(1+\frac{2.327...}{100}\right)^{8}$ **[M1]**

135 000 **[A1]**

[Back to Question] *[Total Marks: 5]*

**6** **(a)** $45000 \times \left(1 - \frac{x}{100}\right)^4 = 26000$ **[M1]**

$1 - \frac{x}{100} = \sqrt[4]{\frac{26000}{45000}}$ or $1 - \frac{x}{100} = 0.8711...$ **[M1]**

12.9 **[A1]**

**(b)** $45000 \times (0.8711...)^n < 10000$ or $(0.8711...)^n < \frac{2}{9}$ **[M1]**

Testing values or using logarithms: $n = 10 \implies \$11342$ and $n = 11 \implies \$9881$ **[M1]**

11 **[A1]**

[Back to Question] *[Total Marks: 6]*

**7** **(a)** $28\,500 \times \left(1 - \frac{12}{100}\right)^5$ **[M1]**

$28\,500 \times 0.88^5$ **[M1]**

15 040 **[A1]**

**(b)** $35\,000 \times \left(1 - \frac{r}{100}\right)^4 = 15\,200$ **[M1]**

$\left(1 - \frac{r}{100}\right)^4 = \frac{152}{350}$ **[M1]**

$1 - \frac{r}{100} = \sqrt[4]{\frac{152}{350}} = 0.8123\ldots$ **[M1]**

18.8 (accept 18.76 to 18.77) **[A1]**

[Back to Question] *[Total Marks: 7]*

**8** **(a)** Let $k = 1 - \frac{r}{100}$. Then $M_0 k^2 = 80$ and $M_0 k^5 = 40.96$ **[M1]**

$k^3 = \frac{40.96}{80} = 0.512 \implies k = \sqrt[3]{0.512} = 0.8$ **[M1]**

$1 - \frac{r}{100} = 0.8 \implies r = 20$ **[A1]**

**(b)** $M_0 \times 0.8^2 = 80$ or $M_0 \times 0.64 = 80$ **[M1]**

125 **[A1]**

**(c)** $125 \times 0.8^n < 10$ or $0.8^n < 0.08$ **[M1]**

Testing values: $n = 11 \implies 0.8^{11} = 0.0858...$ and $n = 12 \implies 0.8^{12} = 0.0687...$ **[M1]**

12 **[A1]**

[Back to Question] *[Total Marks: 8]*

**9** **(a)** $2500 \times \left(1 + \frac{r}{100}\right)^6 = 9100$ **[M1]**

$\left(1 + \frac{r}{100}\right)^6 = 3.64$ **[M1]**

$1 + \frac{r}{100} = 3.64^{\frac{1}{6}} = 1.2403\ldots$ **[M1]**

24.0 (accept 24.03 . . . ) **[A1]**

**(b)** $2500 \times 1.2403^{10}$ or $9100 \times 1.2403^{4}$ **[M1]**

21 500 **[A1]**

**(c)** $2500 \times 1.2403^{n} > 50\,000 \implies 1.2403^{n} > 20$ **[M1]**

$n \times \log(1.2403) > \log(20) \implies n > 13.91\ldots$ (or equivalent trial and error) **[M1]**

14 **[A1]**

**[Back to Question]** *[Total Marks: 9]*

**10** $26000 \times (1 - 0.15)^{4}$ or $26000 \times 0.85^{4}$ **[M2]**

If M0, allow [M1] for $26000 \times 0.85^{k}$ where $k > 1$

13572 **[A1]**

**[Back to Question]** *[Total Marks: 3]*

**11 (a)** $1500 \times 1.08^{12}$ **[M1]**

3777 **[A1]**

**(b)** $1500 \times 1.08^{n} > 3000$ or $1.08^{n} = 2$ **[M1]**

Trial with values (e.g., $1.08^{9} = 1.999...$ and $1.08^{10} = 2.158...$) or uses logarithms (e.g., $n = \frac{\log(2)}{\log(1.08)} = 9.006...$) **[M1]**

10 **[A1]**

**[Back to Question]** *[Total Marks: 5]*

**12 (a)** $\left(1 + \frac{r}{100}\right)^{2} = 1.44$ or $x^{2} = 1.44$ **[M1]**

$1 + \frac{r}{100} = 1.2$ or $x = 1.2$ **[M1]**

20 **[A1]**

**(b)** Number of doubling periods $= 45 \div 15 = 3$ or $80 \times 2^{3}$ **[M1]**

640 **[A1]**

**[Back to Question]** *[Total Marks: 5]*